$$r(t) = \langle \mathcal{E}_-(t) | \mathcal{E}_+(t) \rangle = \prod_i \exp\left[-\frac{1}{2} \left| \lambda_i(t) \right|^2 \right]$$
$$= \exp\left[-\sum_i 4 \frac{\left| g_i \right|^2}{\omega_i^2} \left(1 - \cos \omega_i t \right) \right] . \qquad (5.108)$$